# Hints: Abstract algebra – 6. The hardest problems

Class work

*The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.*

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## Problem 1

**Hint 1.** Do not work with $A$ itself. Choose a non-empty $X \subseteq A$ that minimises the ratio $|X+A|/|X|$, and call the minimum $K$. Then $K \leqslant |A+A|/|A| \leqslant c$, and $|Z+A| \geqslant K|Z|$ for every non-empty $Z \subseteq A$.

**Hint 2.** Key lemma (Petridis): for every finite $C \subseteq G$, $|X+A+C| \leqslant K|X+C|$. Iterating with $C = (k-1)A$, $(k-2)A, \dots$ gives $|X+kA| \leqslant K^k|X|$, hence $|kA| \leqslant |X+kA| \leqslant c^k|A|$.

**Hint 3.** Prove the lemma by induction on $|C|$. Write $C = C' \cup \{\gamma\}$ and $Z = \{x \in X : x+A+\gamma \subseteq X+A+C'\}$. Then $|X+A+C| \leqslant |X+A+C'| + |X+A| - |Z+A| \leqslant K|X+C'| + K|X| - K|Z|$.

**Hint 4.** On the other side, $X+C$ is the disjoint union of $X+C'$ and $(X \setminus W)+\gamma$, where $W = \{x \in X : x+\gamma \in X+C'\} \subseteq Z$. So $|X+C| \geqslant |X+C'| + |X| - |Z|$, which completes the induction.

## Problem 2

**Hint 1.** If the ring is a division ring, it is a finite field $\mathbb{F}_q$ (Wedderburn). Take $n = q-1$: then all non-zero $n$-th powers equal 1, and $1+1=1$ is impossible.

**Hint 2.** If it is not a division ring, there is a non-zero non-unit $a$. Some power $e = a^k$ is idempotent, and $e \neq 1$. If $e \neq 0$, then $x = e$, $y = 1-e$, $z = 1$ solve the equation for every $n$.

**Hint 3.** If $a$ is nilpotent, replace it by a suitable power to get $b \neq 0$ with $b^2 = 0$. For $n \geqslant 2$ take $x = b$, $y = z = 1$; for $n = 1$ take $x = 1$, $y = b$, $z = 1+b$.

## Problem 3

**Hint 1.** Translate the conditions into permutations of $\mathbb{Z}_n$. Let $\tau(x) = x+1$. Then $f$ is a fixed-point-free involution, and the third condition says $(f\tau)^3 = \mathrm{id}$.

**Hint 2.** Compute signs: $f$ is a product of $n/2$ disjoint transpositions (so $n$ is even), and $\tau$ is an $n$-cycle, of sign $-1$.

**Hint 3.** $f\tau$ has order dividing 3, so it is even. Hence $(-1)^{n/2} \cdot (-1) = 1$, i.e. $n/2$ is odd.

## Problem 4

**Hint 1.** $\mathbb{Q}^*$: find a subgroup $H' \leqslant H$ and a monomorphism $H' \to \mathbb{Q}^*$ that cannot be extended because a root is missing in $\mathbb{Q}$. Try $H = \mathbb{Z} \supset H' = 2\mathbb{Z}$ with $\mu'(2k) = 2^k$. An extension would need $\mu(1)^2 = 2$.

**Hint 2.** $(\mathbb{Q}, +)$ is divisible. Extend $\mu'$ one element at a time. Given $\mu$ on $L$ and $h \notin L$: if no multiple of $h$ lies in $L$, set $\mu(h) = 0$; otherwise let $m > 0$ be minimal with $mh \in L$ and set $\mu(h) = \mu(mh)/m$. Check that this is well defined on $L + \mathbb{Z}h$.

**Hint 3.** Finish with Zorn's lemma applied to the extensions of $\mu'$ ordered by inclusion (any homomorphism extending $\mu'$ is fine; it need not be injective).

## Problem 5

**Hint 1.** Show that $H$ always equals $L = \langle \text{all } 2^n \text{ words of length } n \text{ in } g, h \rangle$. Clearly $H \subseteq L$, since $H$ is generated by some of these words. Call a length-$n$ word "good" if it lies in $H$; every length-$n$ factor of $w$ is good.

**Hint 2.** Rectangle rule: if $PR$, $SR$, $SQ$ are good (with $|P| = |S|$), then $PQ = (PR)(SR)^{-1}(SQ)$ is good.

**Hint 3.** Two facts about $w$. Every factor has a right extension. And since $w$ is not eventually periodic, for every length $\ell$ there is a right special factor $C$ of length $\ell$ ($Cg$ and $Ch$ both occur). Otherwise the next letter would be determined by the previous $\ell$ letters, and $w$ would be eventually periodic.

**Hint 4.** Prove by induction on $r$ that every word $BA$ with $B$ a factor of length $n - r$ and $A$ arbitrary of length $r$ is good. For the step, take $BxA$ with $|A| = r$. Choose $y$ with $By$ a factor and $C$ right special of length $n - r - 1$. Then $BxA = (ByA)(CyA)^{-1}(CxA)$. At $r = n$ all words are good.